📌 1. Approaches to Knowledge Representation (10 Marks)

Definition

- **Knowledge representation = way to store information in AI systems for reasoning**

Approaches

1. **Logic-Based Representation**
   - **Uses propositional or predicate logic**
   - **Example: Parent(John, Mary) ∧ Female(Mary)**
2. **Semantic Networks**
   - **Graph structure of nodes (concepts) and edges (relationships)**
   - **Example: “Cat → is a → Mammal”**
3. **Frames**
   - **Structured representation with slots and values**
   - **Example: Hospital frame → {Name, Location, Beds}**
4. **Production Rules**
   - **IF–THEN rules**
   - **Example: IF Fever ∧ Cough → THEN Flu**

Diagram

[Logic] [Semantic Network] [Frames] [Rules]

| | | |

Facts/Rules → Graph/Nodes → Slots/Values → IF-THEN

📌 2. Primary Rules of Inference in Propositional Logic (10 Marks)

Definition

- **Rules used to derive new facts from existing propositions**

Primary Rules

1. **Modus Ponens:**
   - **P ⇒ Q, P → Q**
   - **Example: If Rain → Wet, Rain → Wet**
2. **Modus Tollens:**
   - **P ⇒ Q, ¬Q → ¬P**
3. **Hypothetical Syllogism:**
   - **P ⇒ Q, Q ⇒ R → P ⇒ R**
4. **Disjunctive Syllogism:**
   - **P ∨ Q, ¬P → Q**
5. **Conjunction:**
   - **P, Q → P ∧ Q**
6. **Resolution:**
   - **¬P ∨ Q, P → Q**

Diagram

P ⇒ Q P

↓

Q (Modus Ponens)

📌 3. Forward-Chaining Algorithm in Propositional Logic (10 Marks)

Definition

- **Data-driven reasoning method**
- **Start from known facts, apply rules, infer new facts**

Algorithm

1. **Initialize Fact List with known facts**
2. **Repeat:**
   - **Select a rule whose antecedent is satisfied by facts**
   - **Add consequent to fact list if not already present**
3. **Stop when goal is found or no more rules apply**

Example

- **Facts: Fever, Cough**
- **Rule: Fever ∧ Cough → Flu**
- **Forward chaining deduces Flu**

Diagram

Facts: Fever, Cough

↓

Apply Rule: Fever ∧ Cough → Flu

↓

New Fact: Flu

📌 4. Backward-Chaining Algorithm in Propositional Logic (10 Marks)

Definition

- Goal-driven reasoning
- Start from goal, check if it can be inferred from known facts

Algorithm

1. Identify goal
2. Check if goal is known fact
3. If not, find rules whose consequent = goal
4. Recursively set antecedents as sub-goals
5. Stop when all sub-goals are satisfied or fail

Example

- Goal: Flu
- Rule: Fever $\wedge$ Cough → Flu
- Sub-goal: Check Fever and Cough facts

Diagram

Goal: Flu

↑

Sub-goals: Fever $\wedge$ Cough

↑

Check Facts: Fever? Cough?

📌 5. Resolution Method (Clause Form + Binary Resolution) (10 Marks)

Steps

1. **Convert propositions to CNF (Conjunctive Normal Form)**
   - **Eliminate ⇒, ⇔, move ¬ inside, distribute ∨ over ∧**
2. **Express as clauses**
   - **Each clause = disjunction of literals**
3. **Binary Resolution**
   - **Select pair of clauses containing complementary literals**
   - **Derive new clause until empty clause (□) or goal proven**

Example

- **KB: P ∨ Q, ¬P ∨ R, ¬Q → Goal: R**
- **Resolution:**
  1. **P ∨ Q, ¬Q → P**
  2. **P, ¬P ∨ R → R ✅**

Diagram

Clauses: P∨Q, ¬P∨R, ¬Q

↓

Resolution Step1: P∨Q + ¬Q → P

↓

Resolution Step2: P + ¬P∨R → R

Goal Proven

📌 6. Issues in Knowledge Representation & Inheritance in Semantic Nets (10 Marks)

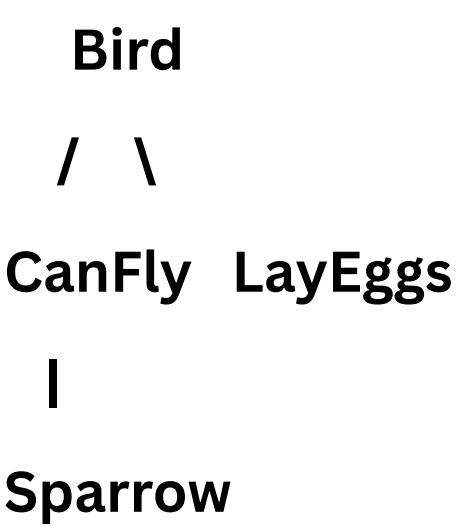
Issues in Knowledge Representation

1. **Expressiveness – Represent all relevant facts clearly**
2. **Efficiency – Reasoning should be fast**
3. **Consistency – Avoid contradictions**
4. **Scalability – Handle large knowledge bases**
5. **Uncertainty Handling – Represent incomplete or uncertain knowledge**

Inheritance in Semantic Net

- **Definition: Concept can inherit properties from parent classes**
- **Example:**
  - **Bird → CanFly = True**
  - **Sparrow is a Bird → Sparrow inherits CanFly = True**

Diagram

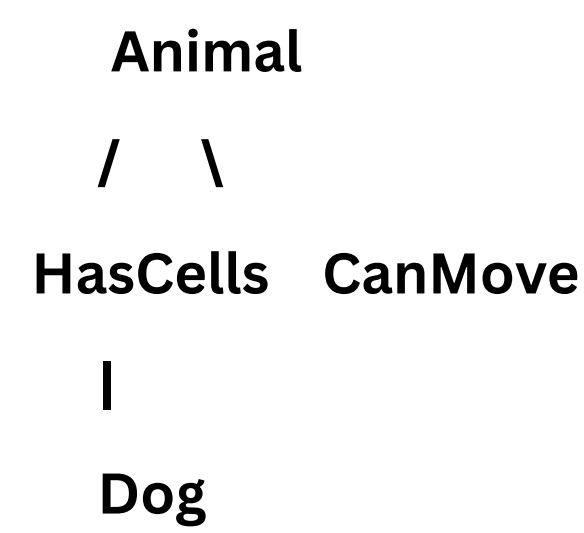


(Inherits CanFly, LayEggs)

📌 7. Explain Inheritance in Semantic Net (10 Marks)

- Semantic Net = Graph of nodes (concepts) and edges (relations)
- Inheritance: A child node automatically gets properties of parent
- Example:
  - Animal → HasCells = True
  - Dog → Animal → Dog inherits HasCells = True

Diagram



(Inherits HasCells, CanMove)

📌 8. Semantic Network Representation: Advantages & Disadvantages (10 Marks)

Definition

- Knowledge represented as nodes (concepts) and links (relations)

Advantages

1. Easy to visualize relationships
2. Supports inheritance

3. **Natural representation of objects & properties**

Disadvantages

1. **Can become complex for large KB**
2. **Difficult to represent procedural knowledge**
3. **Limited formal semantics for logical reasoning**

Diagram

Cat → is a → Mammal → is an → Animal

Cat → has → Fur

📌 9. Knowledge Representation Using Semantic Network (10 Marks)

- **Represents concepts and relationships**
- **Nodes = Objects / Concepts**
- **Links = Relationships / Properties**

Example

- **John → is a → Teacher**
- **Teacher → has → Subject**
- **Subject → Math**

Diagram

John

|

is a

|

Teacher

|

has

|

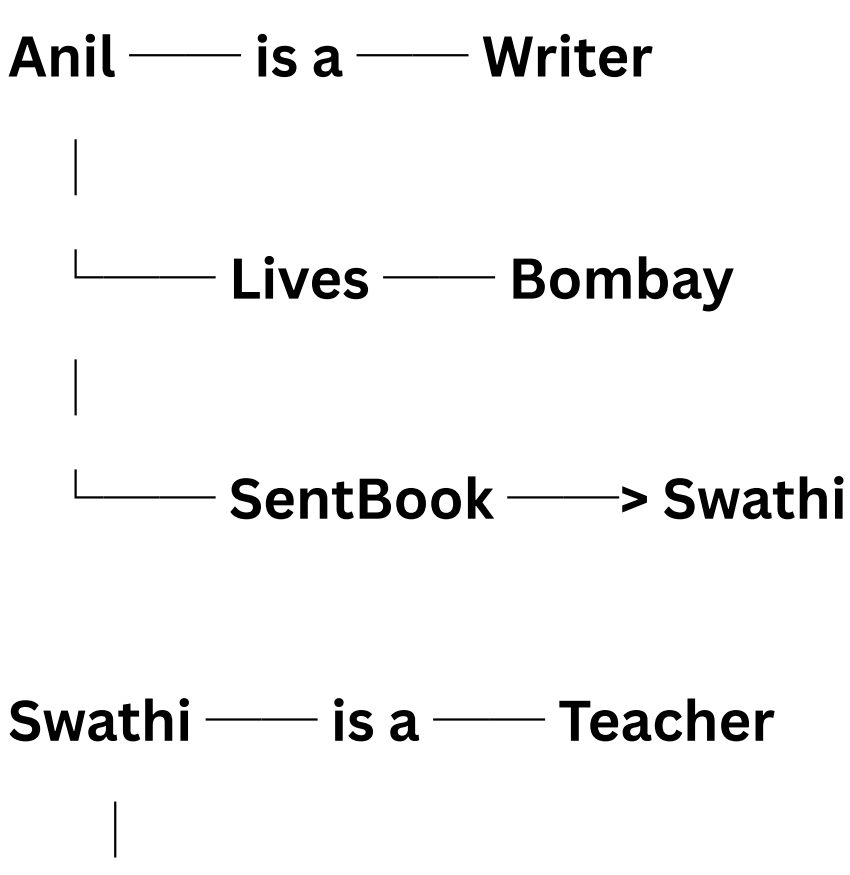
Subject → Math

📌 10. Represent Knowledge Using Semantic Network (Anil & Swathi Example) (10 Marks)

Given Facts

1. **Anil → Writer**
2. **Anil → Lives → Bombay**
3. **Swathi → Teacher**
4. **Swathi → Lives → Bangalore**
5. **Anil → SentBook → Swathi**
6. **Swathi → SentThanks → Anil**

Semantic Network Diagram



└──── Lives ──── Bangalore

│

└──── SentThanks ────> Anil

📌 11. Represent Knowledge Using Semantic Network (Tom/Cat Example)

Facts

- Tom → Cat
- Tom → Caught → Bird
- Tom → OwnedBy → John
- Tom → Color → Ginger
- Cats → Like → Cream
- Cat → On → Mat
- Cat → IsA → Mammal
- Bird → IsA → Animal
- Mammals → IsA → Animals
- Mammals → Has → Fur

Semantic Network Diagram

Tom ──── is a ──── Cat ──── is a ──── Mammal ──── is a ──── Animal

│ │

├── Caught ────> Bird ──── is a ──── Animal

├── OwnedBy ────> John

└── Color ──── Ginger

Cats —— Like —— Cream

Cat —— On —— Mat

Mammals —— Has —— Fur

📌 12. Different Types of Knowledge & Knowledge Representation Cycle

Types of Knowledge

1. **Declarative Knowledge – Facts, concepts (What)**
2. **Procedural Knowledge – How to do tasks (How)**
3. **Semantic Knowledge – Relationships between concepts**
4. **Episodic Knowledge – Specific events / experiences**

Knowledge Representation Cycle

1. **Acquisition – Gather facts/knowledge**
2. **Representation – Encode using logic, frames, semantic nets, rules**
3. **Reasoning – Draw conclusions using inference**
4. **Learning – Update KB with new knowledge**

Diagram

Acquisition → Representation → Reasoning → Learning → Back to Acquisition

📌 13. Extended Semantic Networks (10 Marks)

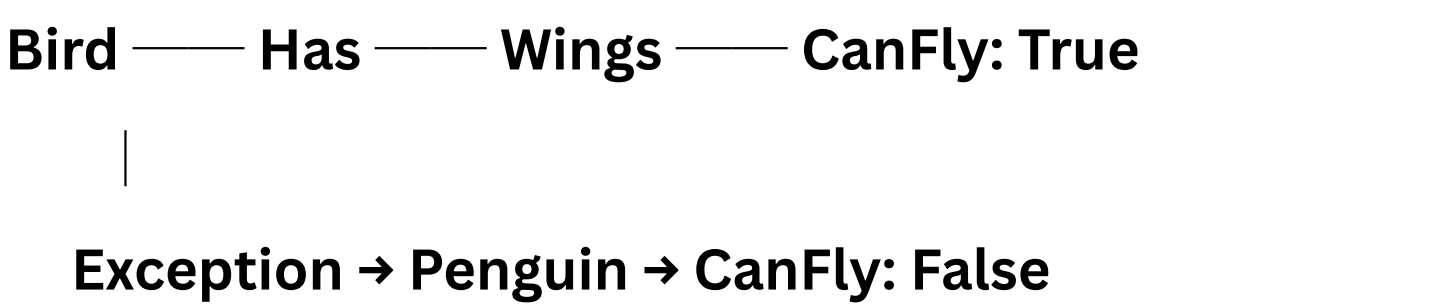
Definition

- Semantic networks with additional expressive power
- Can represent:
  - Complex relationships
  - Quantifiers
  - Procedural knowledge

Example

- “All birds with wings can fly except penguins”
  - Node: Bird
  - Link: Has → Wings
  - Conditional link: CanFly = True (Except → Penguin)

Diagram



📌 14. Extended Semantic Network Example (John gave book to Mary)

Sentence Representation

- John → Gave → Book → To → Mary

Diagram



- Shows action (gave), object (book), recipient (Mary) clearly

📌 15. Propositional Logic (10 Marks)

Definition

- **Logic consisting of atomic propositions combined with logical connectives**
- **Statements are either True or False**

Syntax

1. **Atomic propositions: P, Q, R**
2. **Connectives: ¬, ∧, ∨, ⇒, ⇔**
3. **Well-formed formulas (WFF) using parentheses**

Example

- **P: “It is raining”**
- **Q: “The ground is wet”**
- **Rule: P ⇒ Q**

Diagram

Atomic Propositions: P, Q

Connectives: ¬, ∧, ∨, ⇒, ⇔

Example Formula: P ⇒ Q

📌 16. Semantic Net for “John gave the book to Mary”

Representation

- **Nodes: John, Book, Mary**
- **Relation/Link: Gave, Recipient**

Diagram

John —— Gave ——> Book —— Recipient ——> Mary

- Clearly shows actor (John), action (gave), object (Book), receiver (Mary)

📌 17. Key Issues in Knowledge Representation (10 Marks)

Key Issues

1. **Expressiveness** – Ability to represent all knowledge types
2. **Efficiency** – Reasoning should be fast
3. **Consistency** – Avoid contradictions
4. **Scalability** – Large KB handling
5. **Uncertainty Handling** – Represent incomplete/uncertain knowledge
6. **Understandability** – Easy for humans to read

How Different Methods Address Issues

Method	Issue Addressed
Logic	Expressiveness, Consistency
Semantic Net	Understandability, Inheritance
Frames	Structured knowledge, Defaults
Rules	Procedural knowledge, Automation

📌 18. Syntax and Semantics of Predicate Logic (10 Marks)

Syntax

- Constants: John, Mary
- Variables: x, y
- Predicates: Parent(x, y)
- Functions: Father(x)
- Connectives: $\neg$, $\wedge$, $\vee$, $\Rightarrow$
- Quantifiers: $\forall$, $\exists$

Semantics

- Meaning of formulas in a domain
- Predicate true if relation holds for given objects

Example

- ∀x (Parent(x, Mary) ⇒ Loves(x, Mary))
  - True if every parent of Mary loves Mary

Diagram

Constants: John, Mary

Variables: x, y

Predicates: Parent(x, y), Loves(x, y)

Quantifiers: ∀, ∃

Connectives: ¬, ∧, ∨, ⇒

📌 19. Semantic Networks and Frames with Inheritance (10 Marks)

Semantic Network

- Graph of nodes (concepts) & links (relations)
- Inheritance: Child node inherits parent properties

Frames

- Structured slots and values
- Support default values, inheritance

Example

Semantic Network:

Animal → Has → Fur

Dog → IsA → Animal (inherits Has → Fur)

Frame:

Dog Frame:

IsA: Animal

Fur: Present (inherited)

Sound: Bark

📌 20. Dempster-Shafer Theory & Combining Beliefs (10 Marks)

Definition

- **Method to combine evidence and measure belief/uncertainty**

Given

- **Evidence1: 60% belief in A, 40% in ¬A**
- **Evidence2: 80% belief in A, 20% in ¬A**

Combination

- **Belief(A) = (0.6×0.8) / (0.6×0.8 + 0.4×0.2) = 0.48 / 0.56 ≈ 0.857**
- **Belief(¬A) = 0.143**

Diagram

Evidence1: A=0.6, ¬A=0.4

Evidence2: A=0.8, ¬A=0.2

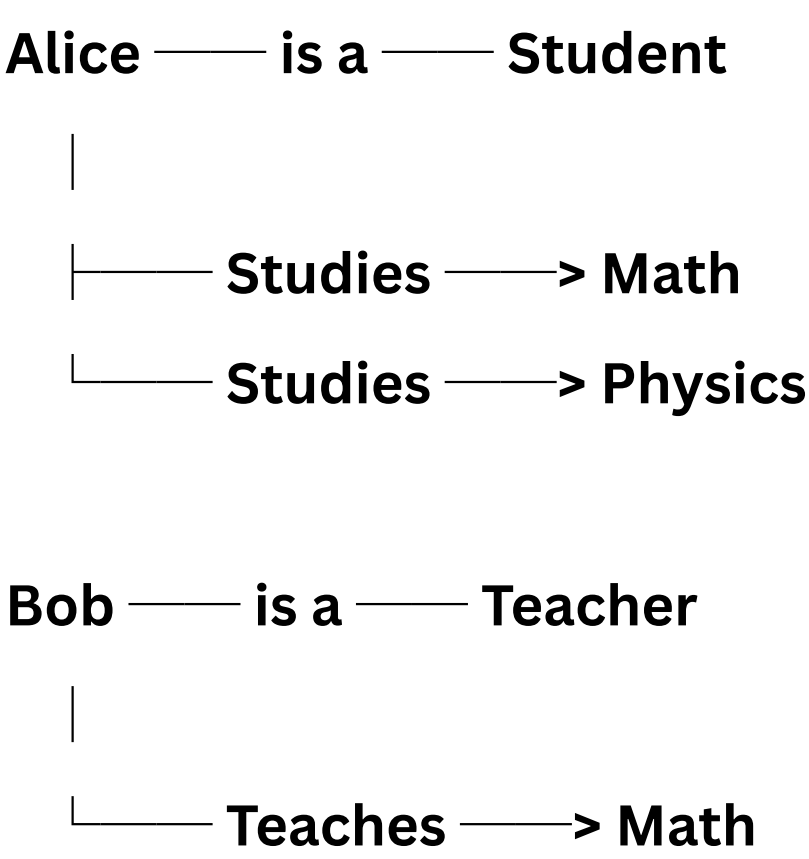
Combine → A ≈ 0.857, ¬A ≈ 0.143

📌 21. Construct a Semantic Network Using Any Knowledge Scenario (10 Marks)

Example Scenario

- **Scenario: “Alice is a student. Alice studies Math and Physics. Bob is a teacher. Bob teaches Math.”**

Semantic Network Representation



- **Nodes = Concepts (Alice, Bob, Math, Physics)**
- **Links = Relations (is a, Studies, Teaches)**

📌 **22. Various Approaches to Knowledge Representation (10 Marks)**

Approaches

1. **Logic-Based Representation**
   - **Propositional & Predicate Logic**
   - **Formal reasoning, precise**
   - **Example: Parent(John, Mary)**
2. **Semantic Networks**
   - **Graph of nodes & links**
   - **Easy visualization, supports inheritance**

3. **Frames**
   - **Structured slots and values**
   - **Default values, inheritance**
   - **Example: Hospital frame → {Name, Location, Beds}**
4. **Production Rules**
   - **IF–THEN rules for reasoning**
   - **Example: IF Fever ∧ Cough → Flu**
5. **Bayesian Networks**
   - **Probabilistic reasoning**
   - **Handles uncertainty**

Diagram

Logic → Deduction

Semantic Net → Nodes & Links

Frames → Slots & Values

Rules → IF-THEN

Bayesian → Probabilities

📌 23. Apply Resolution in Propositional Logic to Prove a Goal (10 Marks)

Steps

1. **Convert KB and goal (¬Goal) to CNF**
2. **Represent as clauses**
3. **Apply binary resolution to derive empty clause (□)**
4. **Goal is proven if empty clause derived**

Example

- KB: P ∨ Q, ¬P ∨ R, ¬Q
- Goal: R
- Resolution:

Step1: P ∨ Q, ¬Q → P

Step2: P, ¬P ∨ R → R ✅

📌 24. Knowledge Representation Using Frames (10 Marks)

Definition

- Frame = data structure to represent stereotypical objects
- Each frame has slots with values, defaults, constraints

Components

1. Slots – Attributes of object
2. Facet – Details about slot (value, default, constraints)
3. Inheritance – Child frame inherits slots from parent

Example

- Frame: Student
  - Name: Alice
  - Age: 20
  - Course: Math
  - Parent: Person frame

Diagram

Student Frame

Name: Alice

Age: 20

Course: Math

IsA: Person

📌 25. Define Frames & Knowledge Representation Using Frames (10 Marks)

- **Frames: Structured way to represent objects with attributes (slots)**
- **Support inheritance, defaults, procedural info**
- **Used in AI for easy knowledge storage and reasoning**

Example

- **Frame: Hospital**
  - **Name: City Hospital**
  - **Location: Mumbai**
  - **Beds: 200**
  - **Departments: Cardiology, Neurology**

Diagram

Hospital Frame

Name: City Hospital

Location: Mumbai

Beds: 200

Departments: Cardiology, Neurology

📌 26. Links in Frames & Hospital Frame Example (10 Marks)

Different Links in Frames

1. **IS-A – Inheritance from parent frame**
   - **Example: Doctor IS-A Person**
2. **Has-A / Slot – Attributes or properties**
   - **Example: Hospital HAS-A Beds**
3. **Default Values – Predefined slot value if not specified**
   - **Example: Beds = 100**
4. **Procedural Attachments – Actions associated with slot**
   - **Example: AdmitPatient → Increment BedsUsed**

Hospital Frame Example

Hospital Frame

Name: City Hospital

Location: Mumbai

Beds: 200

Departments: Cardiology, Neurology

AdmitPatient: Procedure

📌 27. Constraint Propagation in AI (10 Marks)

Definition

- Technique to reduce possible values of variables based on constraints
- Used in knowledge-based reasoning

Role

- Helps in narrowing search space
- Speeds up problem-solving

Example

- Sudoku: Cell values must satisfy row, column, and block constraints
- Possible values get reduced automatically

Diagram

Cell1: {1,2,3,4} → Constraint → Remove 1,3 → {2,4}

Cell2: ...

📌 28. Constraint Propagation in Rule-Based Systems / Semantic Networks (10 Marks)

- Rule-Based Systems
  - IF–THEN rules apply constraints
  - Example: IF Temperature > 100 → Fever = True
- Semantic Networks

- Inheritance or relational constraints propagate values
- Example: Bird → CanFly = True → Sparrow inherits CanFly = True

Diagram

Rule-Based: IF Temp>100 → Fever

Semantic Net: Bird → CanFly → Sparrow

📌 29. Knowledge Representation Using Production Rules (10 Marks)

Definition

- Rules in IF–THEN format
- Represent procedural knowledge

Example

1. IF Fever ∧ Cough → THEN Flu
2. IF StudentScore > 90 → THEN Grade = A

Advantages

- Easy to modify
- Supports automatic reasoning

Diagram

Facts → IF Conditions Met → THEN Actions / Conclusions

📌 30. Working of Rule-Based Deduction System in AI (10 Marks)

Components

1. Knowledge Base – Collection of rules/facts

2. **Inference Engine – Applies rules to derive conclusions**
3. **Working Memory – Stores current facts**
4. **User Interface – Input queries and show results**

Workflow

1. **Input facts into working memory**
2. **Inference engine matches rules → fires applicable rules**
3. **Derive new facts → Update working memory**
4. **Repeat until goal is reached**

Diagram

Knowledge Base + Working Memory

|

v

Inference Engine → Derives New Facts

|

v

User Output

📌 31. Components & Workflow of a Rule-Based Inference Engine (10 Marks)

Components

1. **Working Memory – Stores current facts**
2. **Knowledge Base – Contains IF–THEN rules**

3. **Inference Engine – Applies reasoning methods (forward/backward chaining)**
4. **Pattern Matcher – Matches facts with rule conditions**
5. **Conflict Resolver – Decides which rule to fire when multiple applicable**

Workflow

1. **Initialize working memory with facts**
2. **Match rules using pattern matcher**
3. **Resolve conflicts → Fire rule → Update working memory**
4. **Repeat until goal reached**

Diagram

Facts → Working Memory

Rules → Knowledge Base

|

v

Pattern Matcher → Conflict Resolver → Fire Rule → Update Facts

📌 **32. Bayesian Inference in AI (10 Marks)**

Definition

- **Method of reasoning under uncertainty using Bayes' theorem**

Bayes' Theorem

$$
P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}
$$

- H = Hypothesis, E = Evidence

Usage in AI

- Medical diagnosis, spam filtering, decision-making under uncertainty

Example

- Disease D probability given symptom S
  - P(D|S) = (P(S|D) × P(D)) / P(S)

Diagram

Prior Probability + Evidence → Bayes' Theorem → Posterior Probability

📌 33. Probability Theory in Intelligent Systems (10 Marks)

Role

- Models uncertainty in knowledge
- Supports decision-making under uncertainty
- Helps predict outcomes from incomplete data

Basic Concepts

1. Random Variables – Represent uncertain events
2. Probability Distributions – Likelihood of events
3. Conditional Probability – P(A|B)

Example

- Robot navigating uncertain environment
- P(Obstacle|SensorReading) → Decide next move

Diagram

Uncertain Data → Probability Model → Decision / Prediction

📌 34. Compare Rule-Based Representation vs Semantic Networks & Frames (10 Marks)

Feature	Rule-Based	Semantic Networks	Frames
Knowledge Type	Procedural (IF-THEN)	Declarative (Concepts & Relations)	Declarative + Structured
Reasoning	Forward/Backward Chaining	Inheritance	Inheritance + Defaults
Representation	Rules	Graph of nodes & links	Slots & Values
Example	IF Fever ∧ Cough → Flu	Cat → IsA → Mammal	Dog Frame: Breed, Age, Color

Diagram

Rule-Based → IF-THEN

Semantic Net → Nodes & Links

Frames → Slots & Values